

3

Cell structure

All cells are enclosed in an extremely thin 'skin' called the **cell membrane**. This membrane allows certain substances to move in and out of cells but prevents the movement of others. Consequently it is said to be **semi-permeable**.

The bulk of a cell is made up of a jelly-like substance called **cytoplasm**. Cytoplasm contains tubular and slit-like passageways called the **endoplasmic reticulum**. This reticulum probably allows materials to move quickly throughout the cell. Cytoplasm also contains starch grains, glycogen granules, oil droplets, and crystals of substances to be excreted. These non-living objects are called **inclusions**.

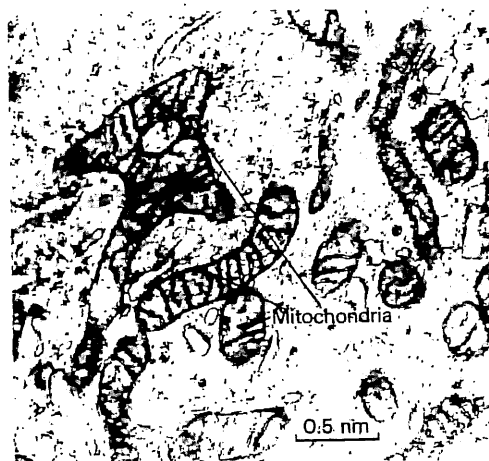
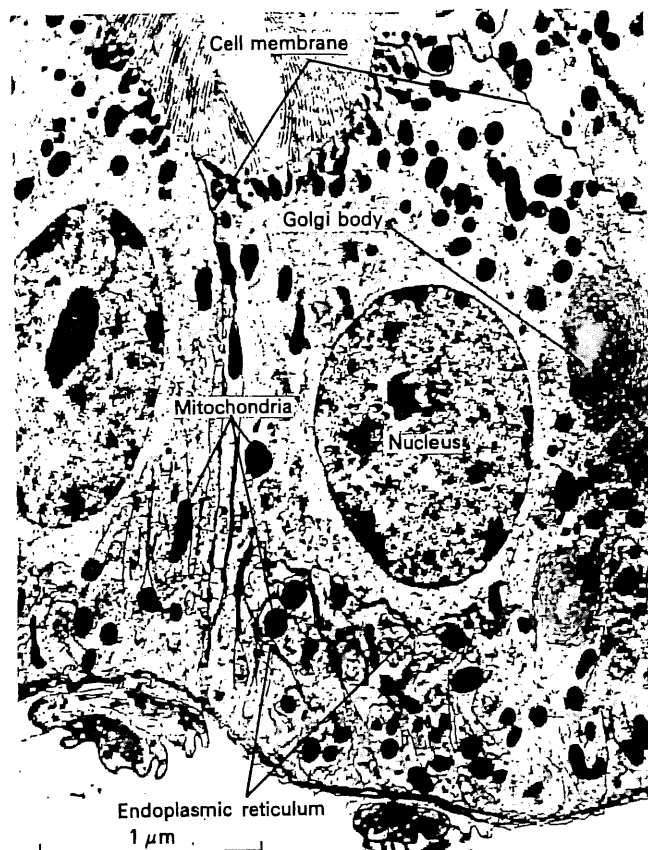
In addition, there are many living objects in cytoplasm called **organelles**. All cells contain round or sausage-shaped organelles called **mitochondria** (*singular* mitochondrion). These are the power plants of a cell. Mitochondria contain enzymes which release energy from food. This process is called **respiration** and is described in Unit 5.

Tiny organelles called **ribosomes** float in the cytoplasm and are attached to the endoplasmic reticulum. Ribosomes contain a chemical called **ribonucleic acid (RNA)**. They take part in the manufacture of proteins, described in Unit 72.

Organelles called **lysosomes** contain enzymes which are released when cells are injured. Lysosomes digest parts of cells and ingested food.

Cytoplasm contains a number of flattened spaces called the **Golgi body**. This is thought to store and concentrate useful substances before they are removed (secreted) from the cell.

All cells contain at least one large spherical or oval object called a **nucleus**. It has a membrane perforated with holes through which chemicals move to and from the cytoplasm, and it is rich in a chemical called **deoxyribonucleic acid (DNA)**. This chemical forms part of objects called **chromosomes**, which only become visible in a nucleus when a cell divides. DNA is concerned with protein manufacture, and plays an essential part in cell division, a process described in Units 67 and 68.



Electron micrographs of cells.

A: A cell from the intestine wall ($1\text{ }\mu\text{m} = 1/1000\text{ mm}$)

B: Mitochondria ($1\text{ }\mu\text{m} = 1/1000\text{ mm}$)

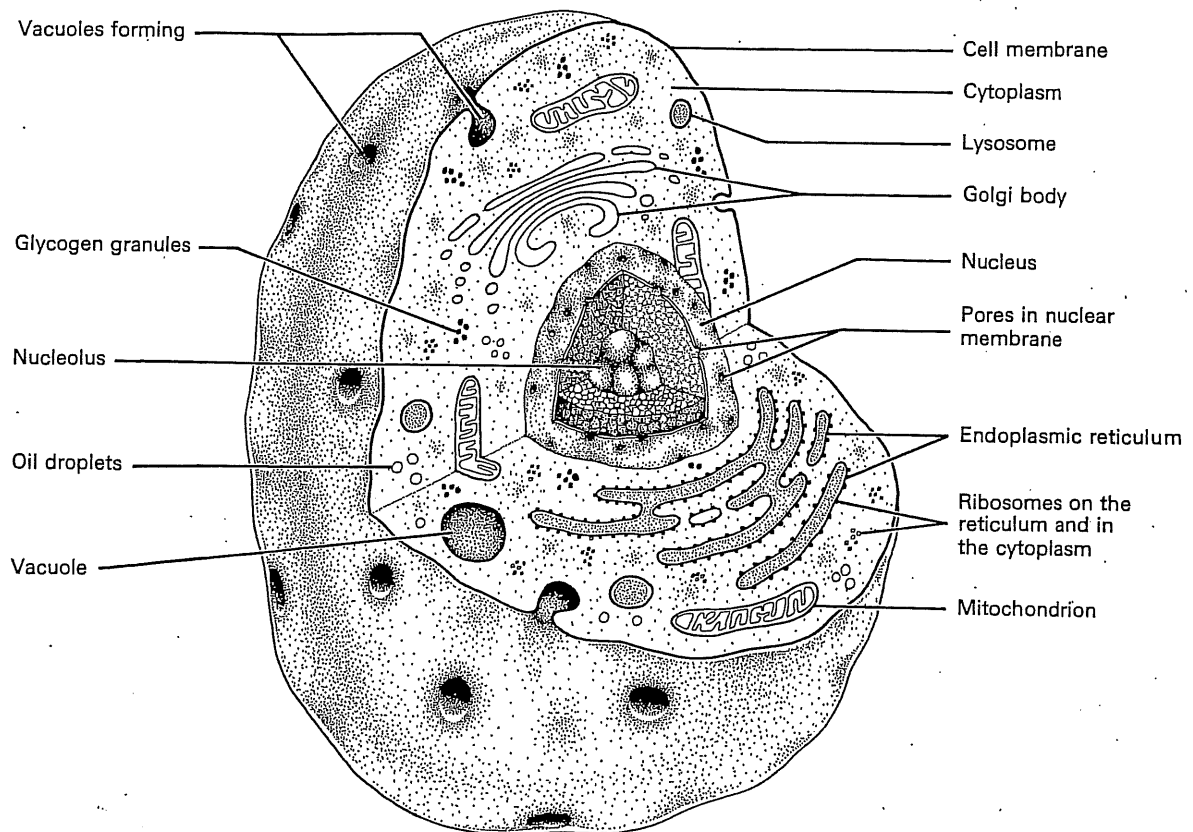


Fig. 1. A greatly magnified animal cell cut open to show its internal structure.

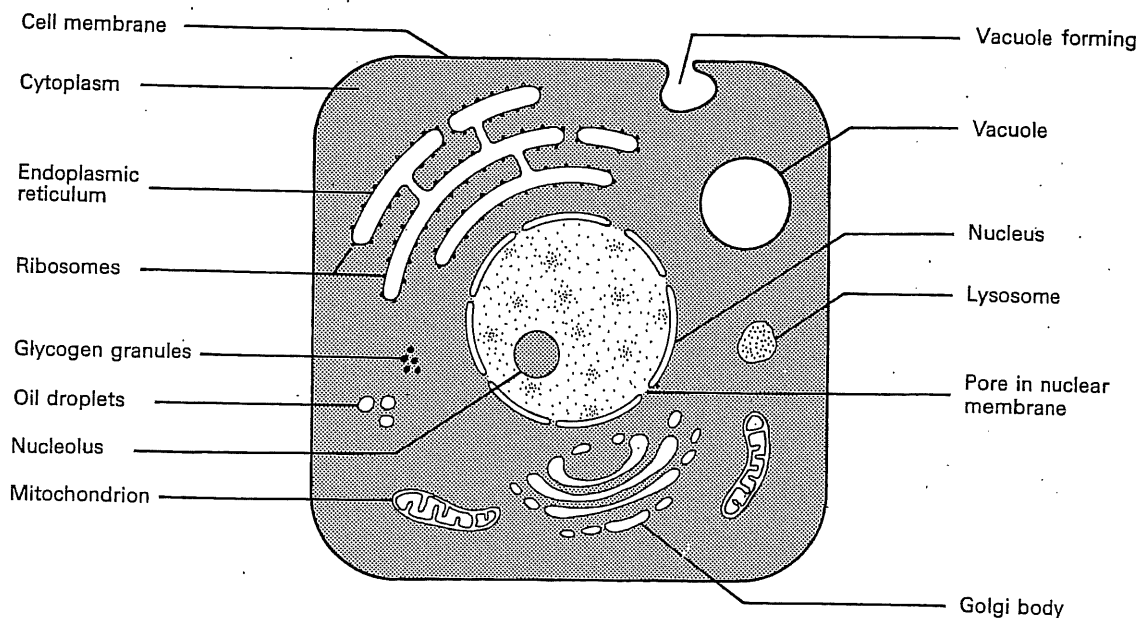


Fig. 2. Diagram of the main parts of an animal cell.

Cell structure

Table 12.1 The parts of a cell

Part	Location	Function
Cell membrane	surrounding the cell	forms the boundary of the cell; controls movement of materials into and out of the cell
Cytoplasm	between the cell membrane and the nucleus	a medium in which the organelles are suspended and in which chemical reactions can occur
Organelles		
Nucleus	often near the centre of the cell	contains the inherited material and controls the cell's activities
Ribosomes	in the cytoplasm, often attached to the endoplasmic reticulum	sites for protein synthesis
Endoplasmic reticulum	penetrating the whole of the cytoplasm	a surface for chemical reactions; transport of materials within the cell; storage of materials
Golgi apparatus	in the cytoplasm, often near the nucleus	concentrates, stores and packages materials for secretion from the cell
Lysosomes	in the cytoplasm	digest material within the cell
Mitochondria	in the cytoplasm	make energy available for cellular activities
Centrioles	in the cytoplasm, usually near the nucleus	involved in the process of cell division
Cilia and flagella	projecting from the main body of the cell	move particles along the cell surface or move the whole cell
Inclusions*	within the cytoplasm	chemical substances used by or stored in the cell

Cells are the basic structural and functional units of plants and animals—the building blocks that make up the organism. The human body contains countless millions of cells. In just 1 mL of human blood there are about five million red blood cells—and there are up to 6000 mL of blood in an adult human—and blood is just one of the many tissues that make up the human body!

The structural organisation of the body

The body is organised on four structural levels. At the lowest level is the cell—the basic unit. Cells are specialised to carry out different functions. Muscle cells are able to shorten in length; red blood cells are able to transport oxygen; cells of mucous membranes secrete mucus; and so on. Cells with similar specialisations that carry out a common function are grouped together into tissues. Tissues are the second structural level. Groups of muscle cells make up muscle tissue, and groups of nerve cells make up nervous tissue.

At the third level of organisation, different types of tissue work together as organs. An organ is normally made up of two or more tissues. The stomach is an organ with lining tissue on the inside and muscular tissue in the wall; the heart contains muscular tissue and nervous tissue.

The highest level of organisation is the system. A system is a group of organs that work together for a common purpose. For example, the role of the respiratory system is to supply the blood with oxygen and to remove carbon dioxide from the blood. The organs that make up the respiratory system are the lungs, diaphragm, intercostal

muscles, bronchi and trachea, larynx, pharynx and nose.

The body systems are all integrated into the one living thing, the **organism** (Fig. 12.1).

Components of cells

Different cells are specialised for different tasks. The shape and structure of cells varies according to the task they are specialised to perform. However, there are certain structures that are common to most cells. Figure 12.2 shows these structures in a diagram of a generalised cell. It is referred to as *generalised* because all cells have in detail and the generalised cell, and they represent any particular cell.

Cells are made up of the following parts (see Table 12.1):

- a **cell membrane**, which surrounds the cell and forms the outer boundary of the cell;
- **cytoplasm**, a thick fluid between the cell membrane and the nucleus;
- **organelles**, structures suspended in the cytoplasm that carry out particular functions;
- **inclusions**, chemical substances occurring as granules or liquid droplets in the cytoplasm.

Cell membrane

The **cell membrane**, or **plasma membrane**, separates the cell from neighbouring cells and separates the cell contents from the external environment. It is very thin—too thin to be seen clearly with a light microscope. The cell membrane is made up of a double layer of lipid molecules (especially phospholipids and

Diffusion

Cell membranes are described as being differentially permeable (see Chapter 3). They allow certain ions and molecules to pass through, but restrict the movement of others. Diffusion occurs because the random movement of ions and molecules distributes them evenly over the available space. If the concentration of certain ions or molecules is greater outside the cell than inside, more particles will move into the cell until the internal and external concentrations are equal. In the lungs, oxygen diffuses through the membranes of the cells forming the capillary walls, and through the membranes of the red blood cells to combine with the haemoglobin inside. At the same time carbon dioxide diffuses in the opposite direction.

Osmosis is a special type of diffusion, involving the net movement of water through a differentially permeable membrane from a higher water concentration to a lower water concentration (see p. 14). Both diffusion and osmosis are described as passive processes, because the cell does not have to use any energy to transfer substances across the membrane.

Active transport

Active transport involves the movement of substances across a cell membrane from a lower to a higher concentration—that is, in the opposite direction from that in which diffusion would occur. Such transport is described as active because the cell must supply energy for the process to occur. In active transport, the molecule or ion to be transported attaches itself to a carrier molecule that takes it across the membrane and releases it on the other side.

Endocytosis and exocytosis

A cell may surround some extracellular material with a fold of the cell membrane. The enfolding membrane then breaks away, and the material is enclosed within the cell in the form of a small bubble of liquid surrounded by a membrane. Such a liquid-filled sac, within the cytoplasm, is called a vesicle. The process by which the vesicle is formed is called **endocytosis** (Fig. 12.4).

Where the material engulfed by the cell includes solid particles, the process is known as **phagocytosis** (cell-eating). If the material taken in is a liquid, the process is called **pinocytosis** (cell-drinking).

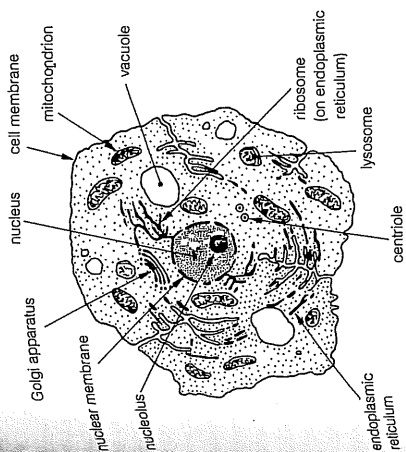


Fig. 12.2 A generalised animal cell showing the components found in many cells

cholesterol) and various proteins associated with the lipid.

Substances that enter or leave the cell must pass through the cell membrane, so the membrane is very important in determining which substances will get into or out of a cell. Materials may pass through a cell membrane in a number of different ways.



Fig. 12.3 Electron micrograph of secretory cells from the lining of the stomach. The large structures are the nuclei, the smaller oval and spherical structures are mitochondria and the rest of the cell contains granular (rough) endoplasmic reticulum

CENTRE FOR HUMAN BIOLOGY, UNIVERSITY OF WESTERN AUSTRALIA

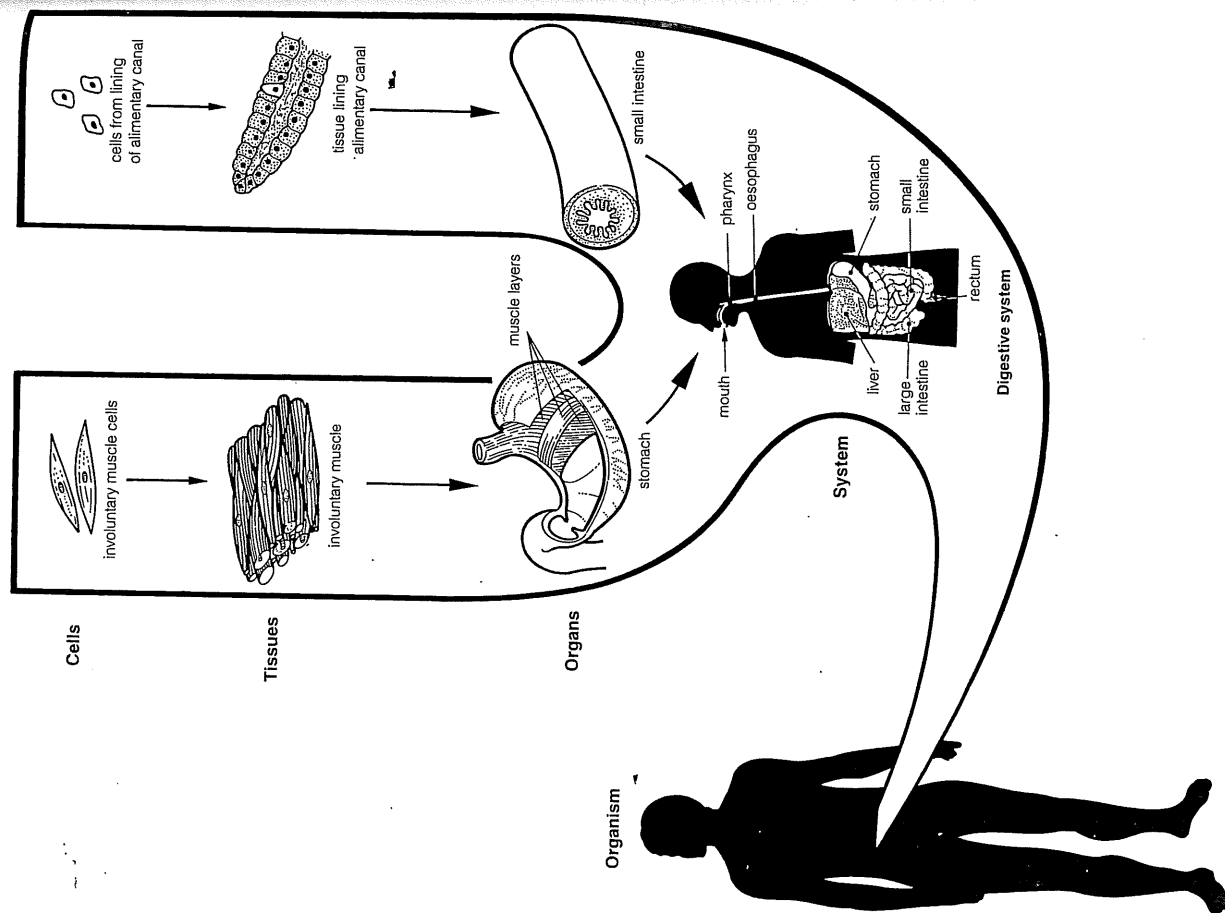


Fig. 12.1 Diagram showing the levels of structural organisation of the human body

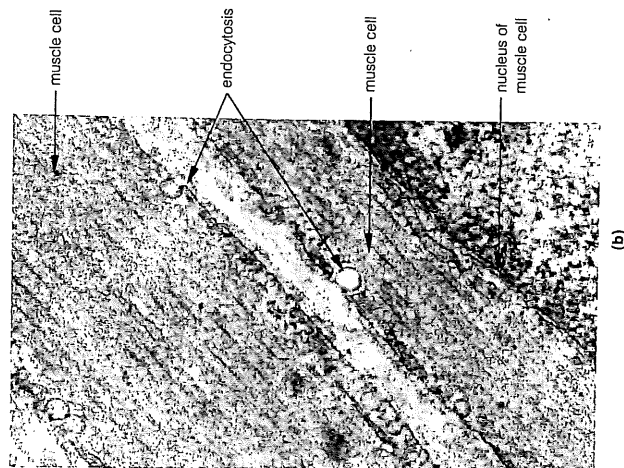
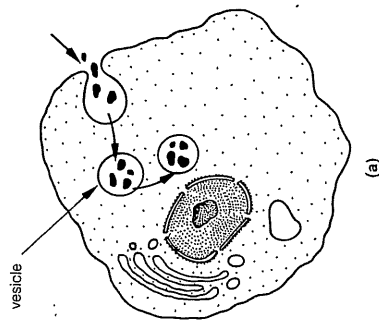


Fig. 12.4 Endocytosis: (a) diagram showing the process; (b) electron micrograph showing endocytosis in two smooth muscle cells (The cells are taking in calcium ions in solution. Is this pinocytosis or phagocytosis?)

CENTRE FOR HUMAN BIOLOGY, UNIVERSITY OF WESTERN AUSTRALIA

If the contents of a vesicle are pushed through the cell membrane, the process is called **exocytosis**. The membrane around the vesicle fuses with the cell membrane and the vesicle contents are passed to the exterior (Fig. 12.5).

Cytoplasm

The **cytoplasm** is the material in which the cell contents are suspended. It fills all the space between the nucleus and the cell membrane. Cytoplasm is 75–90 per cent water, with substances dissolved and suspended in it. Inorganic materials and most carbohydrates are dissolved in the cytoplasm. Other organic compounds, such as proteins and lipids, do not dissolve but are suspended in the watery medium. The term **protoplasm** refers to the nucleus and cytoplasm together—that is, all the contents of the cell.

Organelles

The structures within a cell are called **organelles**. There are a number of different types of organelle and each is specialised for a particular function.

Nucleus

Almost all cells contain a single nucleus, although some, such as liver cells, have two or more nuclei, and mature red blood cells have no nucleus. The nucleus is the largest organelle in the cell and is usually oval or spherical in shape. A **nuclear membrane** separates the nucleus from the

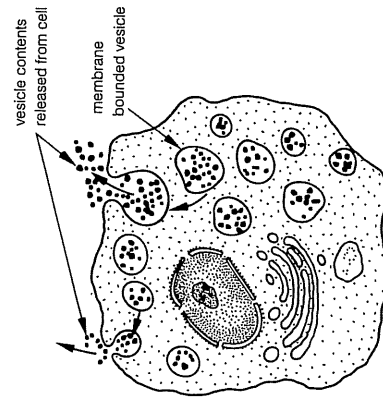


Fig. 12.5 The process of exocytosis

cytoplasm. The nuclear membrane is actually a double membrane—two membranes separated by a space (Fig. 12.6). There are numerous gaps, or **nuclear pores**, in the nuclear membrane. Substances are able to enter and leave the nucleus through the pores.



Fig. 12.6 Electron micrograph (a) and diagram (b) of cell nucleus, showing double membrane and nuclear pores (original magnification 20 000X)

CENTRE FOR HUMAN BIOLOGY, UNIVERSITY OF WESTERN AUSTRALIA

Within the nucleus is the genetic material, mostly DNA (deoxyribonucleic acid), which contains inherited information. When the cell is not dividing, the DNA is in the form of long strands known as **chromatin**. In a dividing cell the threads of chromatin thicken and coil to form **chromosomes**. DNA contains the information that determines the type of proteins a cell can make. As enzymes are proteins, and as enzymes control chemical reactions, this means that the nucleus, with its DNA, determines which chemical reactions will occur in the cell. In this way the nucleus controls the way in which the cell functions.

Within the nucleus is an area called the **nucleolus** (see Fig. 12.2). The nucleolus is composed mainly of RNA (ribonucleic acid), which has a role in protein synthesis. Both the chromatin threads and nucleolus are suspended in a jelly-like material, the **nucleoplasm**.

Ribosomes

Ribosomes are very small, spherical organelles (see Fig. 12.7). At the ribosomes, amino acids are joined together in a sequence determined by the code in the DNA molecules in the nucleus. In this way proteins and polypeptides are formed. (The synthesis of proteins will be described in more detail in Chapter 13.) Ribosomes may be free in the cytoplasm or may be attached to membranes within the cells.

Endoplasmic reticulum

Pairs of parallel membranes extend through the cytoplasm of the cell and connect with the cell membrane and the nuclear membrane. The network of channels formed by the parallel membranes is called the **endoplasmic reticulum**, or ER (Fig. 12.7). It is thought that the membranes of the endoplasmic reticulum provide a surface for chemical reactions, while the channels are for storing or transporting molecules.

Ribosomes are attached to the outside of some membranes of the endoplasmic reticulum (Fig. 12.7). This part is called the **rough**, or **granular**, **endoplasmic reticulum**. Parts of the membranes that do not have any attached ribosomes are referred to as the **smooth**, or **agranular**, **endoplasmic reticulum**.

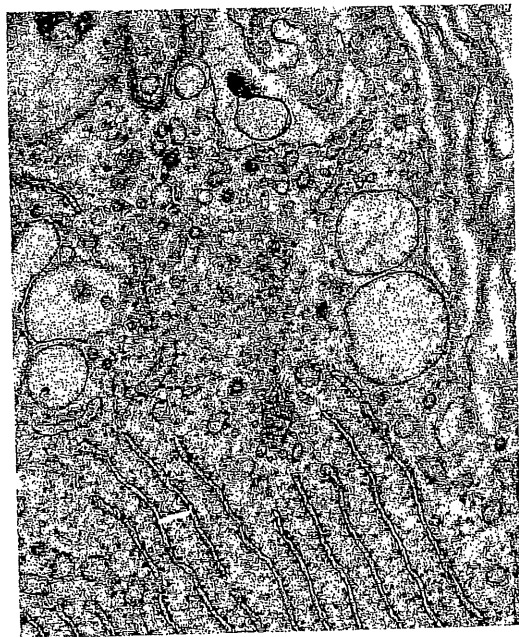
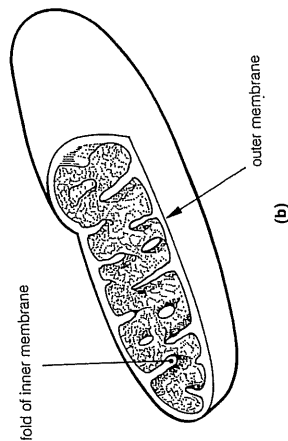


Fig. 12.9 Electron micrograph showing two lysosomes within the cytoplasm of a cell
BIOLOGY, UNIVERSITY OF WESTERN AUSTRALIA

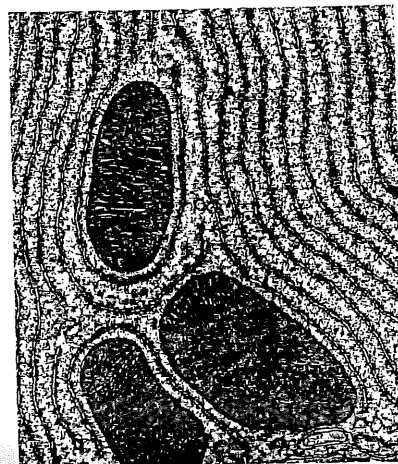
Mitochondria (singular, **mitochondrion**) are spherical or sausage-shaped structures that are spread throughout the cytoplasm. Each has a double membrane. The smooth outer membrane surrounds the mitochondrion. The inner is

arranged into a series of folds that extend into the interior of the organelle (Fig. 12.10). Enzymes involved in cellular reactions that release energy are located on the membranes. Folding of the inner membrane produces a large surface area on which these chemical reactions can take place. Because the reactions of the mitochondria make energy available for the cell's activities, these organelles are often called the 'powerhouses' of the cell. (Details of the reactions at the mitochondria are given in Chapter 13.)



(b)

Fig. 12.10 (a) Electron micrograph showing three mitochondria in section (original magnification 40 000X); (b) diagram showing a three-dimensional view of a mitochondrion
BIOLOGY, UNIVERSITY OF WESTERN AUSTRALIA

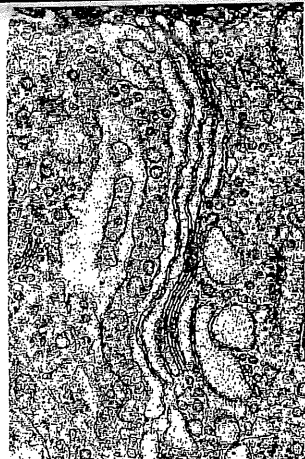


(a)

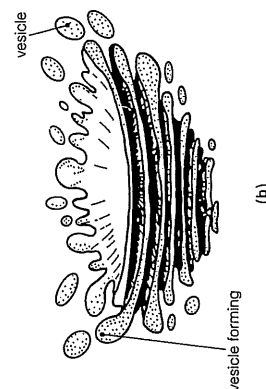
other molecules. At the edges of the membranous bags of the Golgi apparatus the membrane pinches off to form a vesicle containing some of the proteins (Fig. 12.8). These vesicles migrate through the cytoplasm to the cell membrane where the proteins leave the cell by exocytosis.

Lysosomes

Lysosomes are small spheres, bounded by a membrane, that are formed from the Golgi apparatus (Fig. 12.9). They contain digestive enzymes that are able to break down proteins, lipids, nucleic acids and some carbohydrates. When particles or liquids are taken into a cell by endocytosis, they form vesicles in the cytoplasm. Lysosomes can fuse with these vesicles, and the digestive enzymes they contain break down the material inside the vesicle.

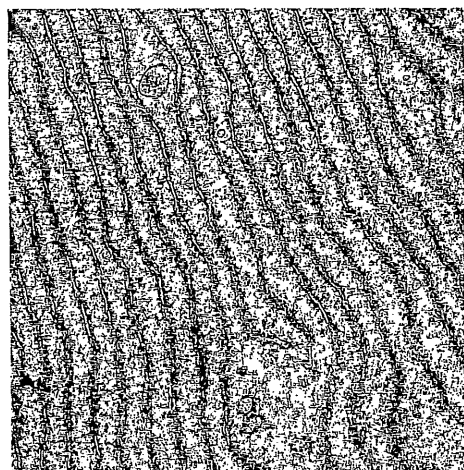


(a)

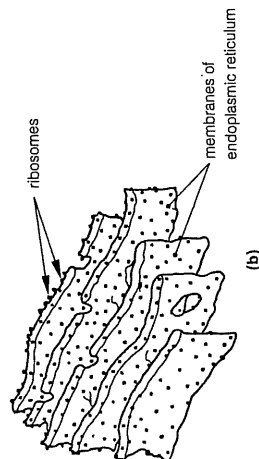


(b)

Fig. 12.8 Electron micrograph (a) showing section through a Golgi apparatus (original magnification 40 000X); (b) diagram indicating the three-dimensional shape of the Golgi apparatus
BIOLOGY, UNIVERSITY OF WESTERN AUSTRALIA



(a)



(b)

Fig. 12.7 Electron micrograph (a) showing granular (rough) endoplasmic reticulum (original magnification 40 000X); (b) diagram indicating the three-dimensional arrangement of the membranes
BIOLOGY, UNIVERSITY OF WESTERN AUSTRALIA

Golgi apparatus

The Golgi apparatus is a series of flattened, membranous bags stacked one upon the other. The ends of the bags are often slightly expanded. Usually the Golgi apparatus is positioned near the nucleus. Its function is to modify proteins and to package them for secretion from the cell. Proteins produced at the ribosomes pass through the channels of the endoplasmic reticulum to the Golgi apparatus. Here they may be modified by the addition of sulfate, phosphate, fatty acid or

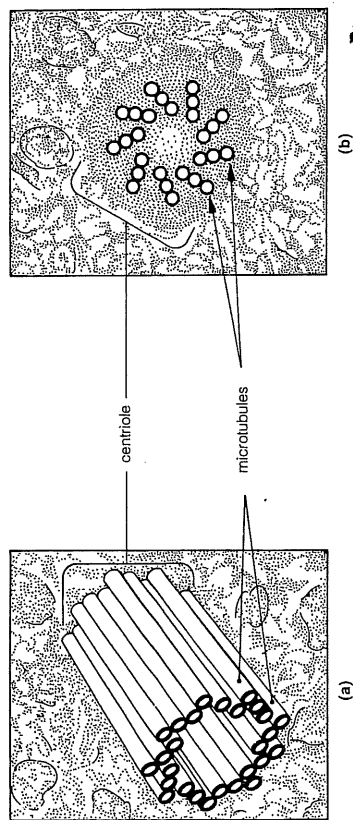


Fig. 12.11 Centrioles: (a) three-dimensional view; (b) in transverse section

Centrioles

The centrioles are a pair of cylindrical structures located near the nucleus. Each cylinder is made up of a ring of very fine tubules, and the pair are arranged at right angles to each other (Fig. 12.11). The centrioles are involved in the reproduction of the cell.

Cilia and flagella

Some cells have fine projections which, by rhythmic movement, can move the whole cell or can move substances over the surface of the cell. If the projections are short and numerous, resembling tiny hairs, they are called cilia (Fig. 12.13). If they are longer, and there are only one or two of them, they are called flagella (Fig. 12.12). One place in which cilia occur is in the cells lining the respiratory tract, where they move mucus and trapped particles towards the throat. In humans, only one type of cell, the sperm cell, has a flagellum; this enables the sperm to swim to the egg.

The structure of cilia and flagella is basically the same. Both consist of a membranous sheath, which is an extension of the cell membrane; inside the sheath are a series of very fine tubules.

Inclusions

Inclusions are chemical substances that are not part of the cell structure but are found within the cytoplasm of the cell. In red blood cells, haemoglobin is an inclusion. In cells of the skin,

hair and iris of the eye, particles of the pigment melanin are inclusions. Other cells may contain inclusions of glycogen, lipids or mucus. If the inclusion is a liquid that is able to mix with the cytoplasm of the cell, it is usually surrounded by a membrane. The inclusion is then called a vacuole.

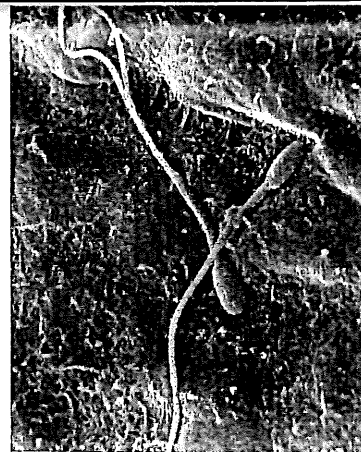


Fig. 12.12 Scanning electron micrograph showing two sperm cells with flagella (note that the flagella of the sperm on the right is forked; a significant proportion of sperm is malformed) CENTRE FOR HUMAN BIOLOGY, UNIVERSITY OF WESTERN AUSTRALIA

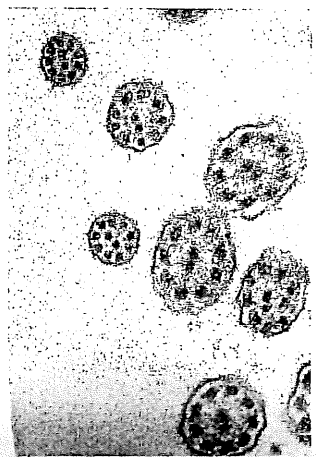
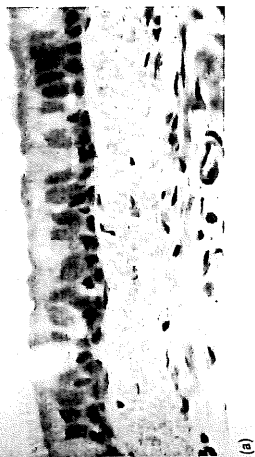


Fig. 12.13 Cilia: (a) cells from the lining of the trachea showing cilia; (b) electron micrograph, showing transverse section (original magnification 50 000X); each cilium has nine pairs of tubules arranged in a circle and two pairs in the centre; (c) electron micrograph, showing longitudinal section (the tubules can be seen inside the cilia) (original magnification 50 000X) CENTRE FOR HUMAN BIOLOGY, UNIVERSITY OF WESTERN AUSTRALIA

Name _____

Cells and Body Organisation

1. What does unicellular mean: _____
2. What does multicellular mean? _____
3. What order should these words go in to show the body organisation?

Cells, Tissues, Organelles, Organ systems, Organism, Organ

_____ → _____ → _____ → _____ →
_____ → _____

4. Organelles are _____ structures performing _____ jobs
5. The mitochondria provides _____ to the cell
6. The ribosomes are involved in _____
7. List 3 types of specialised cells

8. Name one type of tissue in humans _____
9. Name 3 organs in humans _____
10. Are these 'levels of organisation' found in:

- a) Multicellular organisms
- b) Unicellular organisms
- c) both

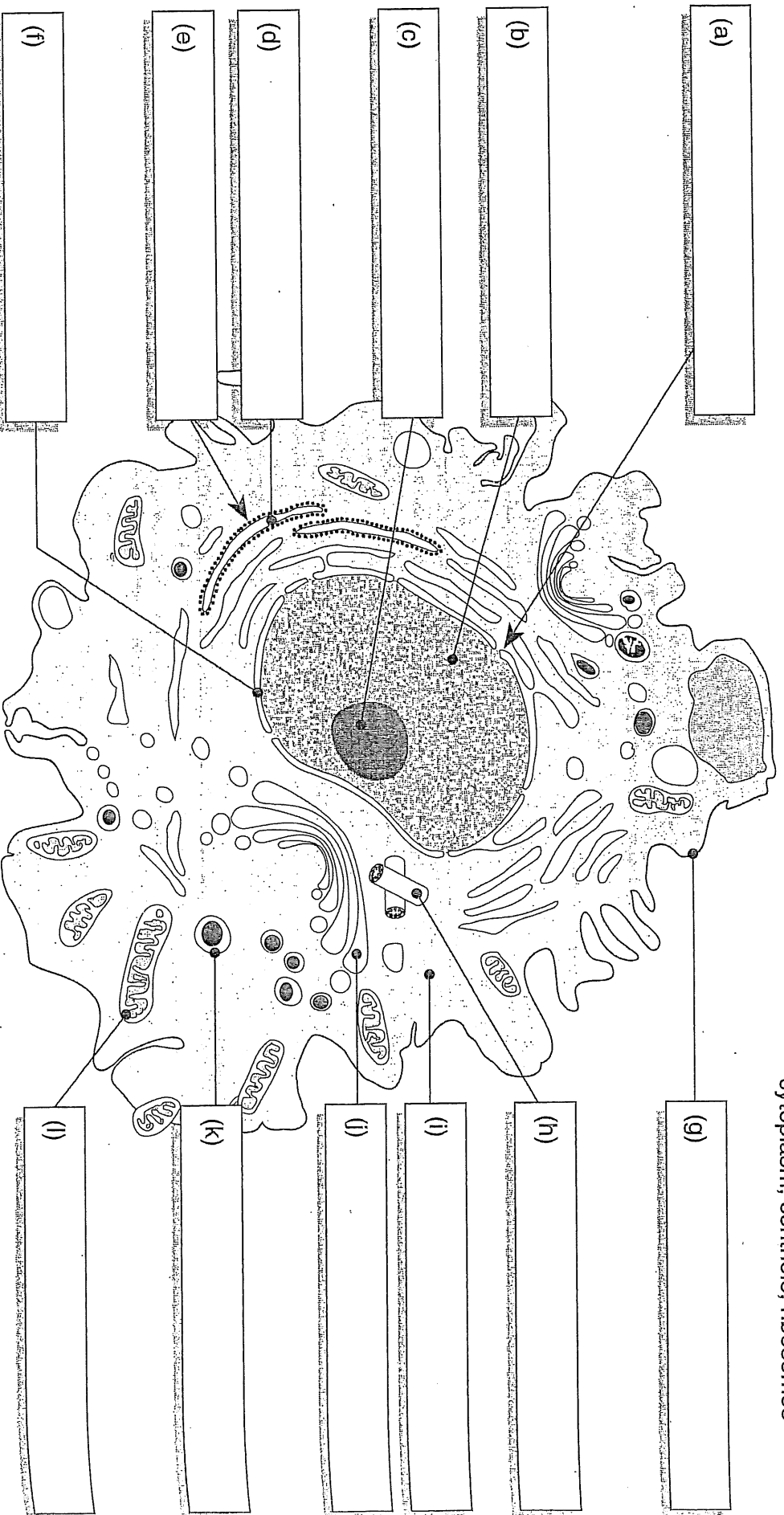
2. Fill in the table with relevant information from the slides.

Part of city	Corresponding cell organelle	Function	Where is it found?	
			Animal	Plant
 City hall				
 Transport system				
 Environment				
 Factory				
 City police				
 Power station				
 Post office				
 Recycling facility				
 Water plant				
 City wall				
 Solar power plant				

Generalised Animal Cell

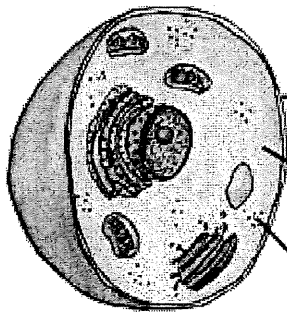
WORD LIST

Lysosome, mitochondrion, plasma membrane, nucleus, nucleolus, nuclear membrane, nuclear pore, Golgi apparatus, endoplasmic reticulum, cytoplasm, centriole, ribosomes



Organelles Revision Worksheet

Name: _____

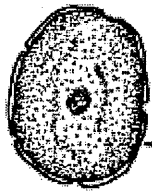


Animal Cell Organelles Quiz

1. Each cell has a protective, semi-permeable, outer layer called the _____.

2. Inside the cell, the watery medium in which all the organelles float, is called _____.

3. Little grains floating around inside the cell involved in proteins production are called _____.



4. The cell _____ contains our **DNA** with all our genetic information found on 23 paired structures called _____ chromosomes.



6. _____ is a series of folded membrane pathways spotted with ribosomes. Together they make new proteins and membranes.



7. _____ has no ribosomes on it and forms containers called **transport vesicles** that are used to move things around inside the cell.



8. _____ are package things to be transported out and around the cell.



9. _____ are vesicles with digestive enzymes inside to break down the things the cell no longer needs.

10. _____ are membrane large membranous sacs for storing things. **Vesicles** are smaller sacs.



11. _____ have a double membrane with finger-like _____ and break down sugars to make energy for the cell.

Word Bank:

Chloroplast
Cell membrane
Cell Wall
Cytoplasm
Nucleus
Smooth endoplasmic reticulum
Mitochondria
Rough endoplasmic reticulum
Golgi apparatus
Lysosomes
Ribosomes

©Sheri Amsel

www.exploringnature.org

Plant cell organelles:

- 12. Photosynthesis happens in the _____
- 13. Plants have an extra rigid structure outside the cell membrane called the _____

11

Active & Passive Transport

Cross-Curricular Focus: Life Science

All living things are made up of tiny units called cells. The cells are surrounded by a covering called a **membrane**. The membrane controls what moves in and out of the cell.

Cells need many kinds of materials in order to thrive. They must have water, oxygen, glucose (sugar), sodium, potassium and a variety of other minerals. The body has to have a delivery system to get these materials where they need to go. In addition, every cell produces waste materials that it needs to get rid of. The body has to have a method of waste collection and disposal. Like a fleet of trucks form a transport system to the pick up and deliver goods in the world, the body has a transport system, too. There are two types of **transport**: **active transport**, and **passive transport**.

Passive transport is easiest for the cells because they don't need to use any energy to make it happen. Diffusion is the simplest and most common form of passive transport. During the diffusion process, tiny particles of the materials that need to be delivered to the cells are spread through either a gas, like oxygen, or a liquid, like water. Areas with high concentrations of the materials slowly diffuse them to areas of low concentrations of the materials. Osmosis is a unique form of passive transport that allows water to pass through the cell membrane but keeps most other materials out.

Active transport is used when the concentration of materials inside the cell is high, and the cell needs to push materials in to areas of high concentration of the materials. Unfortunately, this kind of transport requires the cell to work against the natural direction of diffusion. This causes the cell use energy. One example of this is how nerve cells work. They pump sodium out and pull potassium in. Although they must expend energy in this process, doing so allows them to thrive.

Name: _____

Answer the following questions based on the reading passage. Don't forget to go back to the passage whenever necessary to find or confirm your answers.

1) What is the covering around a cell called?

2) How are active and passive transport different?

3) What is osmosis?

4) What comparison is made to help the reader of this passage better understand how materials are moved throughout the body?

5) Name one type of cell that uses active transport.

I CAN explain how materials move in and out of a cell.

Name _____
Period _____ Date _____

CELL TRANSPORT

CHAPTER 4 Section 1

Match the definition on the left with the term on the right.

1. _____ release of wastes or cell products from inside to outside a cell
2. _____ diffusion of water molecules through a selectively permeable membrane
3. _____ continuous movement of particles but no overall change in concentration
4. _____ movement of particles from an area of higher concentration to one of lower concentration

- a. diffusion
- b. dynamic equilibrium
- c. exocytosis
- d. osmosis

In the space at the left, write true if the statement is true. If the statement is false, change the italicized term to make the statement true. Write this answer in the blank provided.

- _____ 5. In *passive transport*, the movement of particles across a membrane requires energy.
- _____ 6. *Endocytosis* is a process by which a cell membrane surrounds and takes in material from the environment.
- _____ 7. A membrane that allows only some materials to pass through shows *selective permeability*.

Circle the word or phrase that best completes the statement or answers the question.

8. The structure most responsible for maintaining cell *homeostasis* is the
 cytoplasm cell wall mitochondria cell membrane
9. A cell membrane is made up of a(n)
 cholesterol layer enzyme layer lipid bilayer protein layer
10. Which of the following is not a form of passive transport?
 diffusion endocytosis osmosis
11. Diffusion continues until
 equilibrium is reached turgor pressure is reached one side has more
12. If a cell is placed in salt water, water leaves the cell by
 osmosis diffusion active transport phagocytosis
13. A cell moves particles from a region of lesser concentration to a region of higher concentration by
 diffusion osmosis passive transport active transport

Membrane Diffusion Lab

In this investigation, you will explore the process of diffusion across a semi-permeable cell membrane.

Part 1:

1. Go to the following website: <http://phet.colorado.edu/en/simulation/membrane-channels>
2. Click "Run Now" (bright green)
3. Add 5 blue "Gated Channels" – space them out evenly
 - a. Make sure the channels are closed
4. Let the green dots represent solute (like salt) and the blue diamonds represent water molecules
5. Add 20 green dots and 20 blue diamonds into the top section of the cell (let this be the inside of the cell)
6. Add 40 blue diamonds to the bottom section of the cell (let this be the outside of the cell)
7. Is the cell "hypotonic", "hypertonic" or "isotonic" to the surrounding fluid? Explain.

8. You are about to open both of the channels at the same time, what do you expect to happen?

9. Open both types of channels at the same time and begin a timer for 15 seconds.
10. After 15 seconds, pause the simulation and count how many green dots and blue diamonds are on each side of the cell membrane.

	Inside of Cell (top of simulation)		Outside of cell (bottom of simulation)	
	Solute (green dots)	Water molecules (blue diamonds)	Solute (green dots)	Water molecules (blue diamonds)
0 seconds	20	20	0	40
15 Seconds				
30 Seconds				
45 Seconds				
60 Seconds				

11. If you could see this cell and could watch this happen, what would you see the cell do?

Part 2:

12. Click the "Clear Particles" icon
13. Add 20 green dots and 40 blue diamonds into the top section of the cell (let this be the inside of the cell)
14. Add 20 blue diamonds to the bottom section of the cell (let this be the outside of the cell)
15. Is the cell "hypotonic", "hypertonic" or "isotonic" to the surrounding fluid? Explain.

16. You are about to open both of the channels at the same time, what do you expect to happen?

17. Open both types of channels at the same time and begin a timer for 15 seconds.
18. After 15 seconds, pause the simulation and count how many green dots and blue diamonds are on each side of the cell membrane.

	Inside of Cell (top of simulation)		Outside of cell (bottom of simulation)	
	Solute (green dots)	Water molecules (blue diamonds)	Solute (green dots)	Water molecules (blue diamonds)
0 seconds	20	40	0	20
15 Seconds				
30 Seconds				
45 Seconds				
60 Seconds				

19. If you could see this cell and could watch this happen, what would you see the cell do?

Part 3:

20. Click the "Clear Particles" icon
21. Add 20 green dots and 40 blue diamonds into the top section of the cell (let this be the inside of the cell)
22. Add 40 blue diamonds to the bottom section of the cell (let this be the outside of the cell)
23. Is the cell "hypotonic", "hypertonic" or "isotonic" to the surrounding fluid? Explain.

24. You are about to open both of the channels at the same time, what do you expect to happen?

25. Open both types of channels at the same time and begin a timer for 15 seconds.
26. After 15 seconds, pause the simulation and count how many green dots and blue diamonds are on each side of the cell membrane.

	Inside of Cell (top of simulation)		Outside of cell (bottom of simulation)	
	Solute (green dots)	Water molecules (blue diamonds)	Solute (green dots)	Water molecules (blue diamonds)
0 seconds	20	40	0	40
15 Seconds				
30 Seconds				
45 Seconds				
60 Seconds				

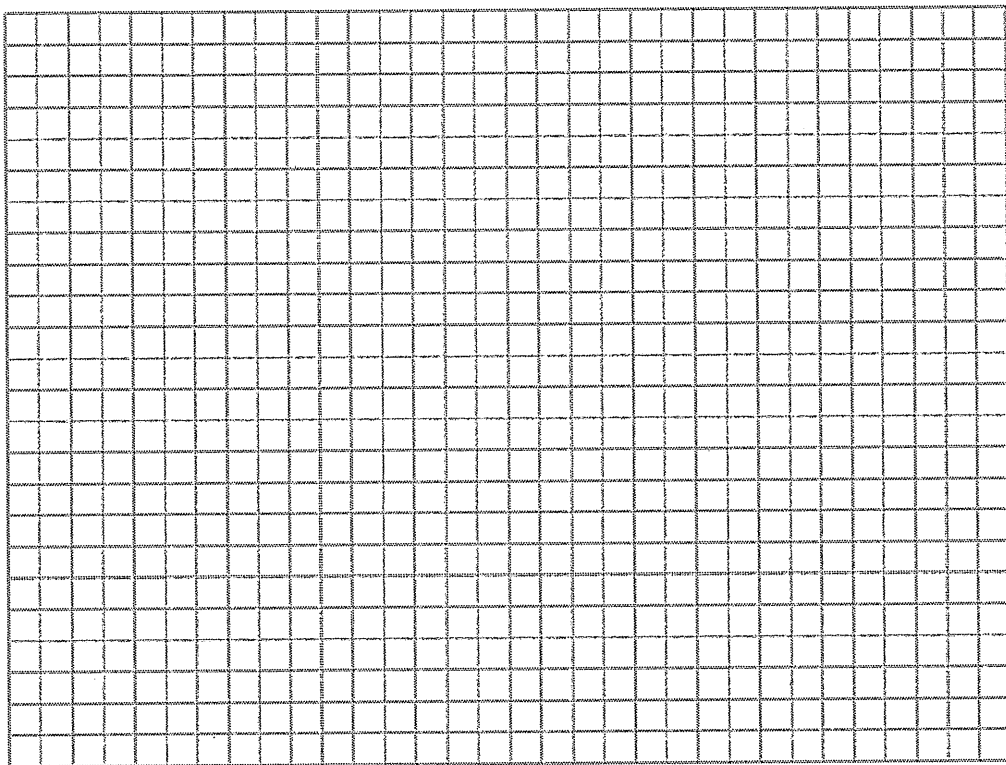
27. If you could see this cell and could watch this happen, what would you see the cell do?

28. Construct a graph. But, before you do, answer the following questions:

a. What do you think is the important information to graph?

b. Is the data you chose to graph, relevant to our lab?

c. What type of graph should you create (i.e. line graph, bar graph, pie chart, etc.)? Why?



29. What can you learn by looking at your graph?

30. Use your graph to make a claim about this investigation.
